Written exam December 19, 2016.

Course name: Network Optimization **Course no:** 42115

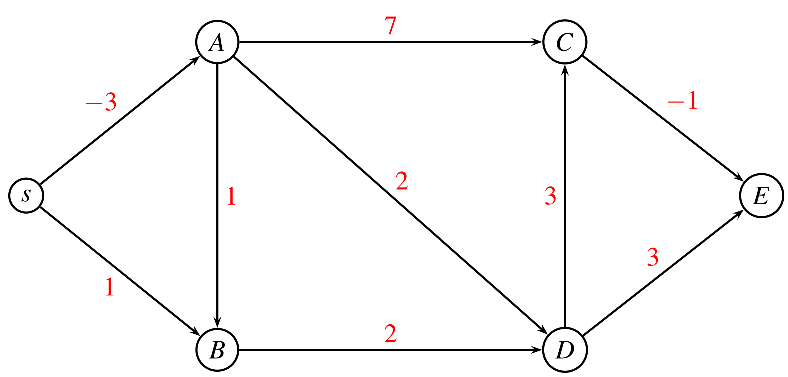
Rules: You are allowed to use your books, notes, slides, computer (but no internet access). You can write in Danish or English.

Assignment: The assignment consists of 12 problems. In some of the problems it is written what an answer is expected to contain. Notice that all figures can be found on the last pages of this assignment. You can use them in your answer by drawing on the figure.

Points: Small problems give 2 points, medium problems give 3 points, large problems 4 points. A total of 35 points can be obtained.

Shortest path

In the following network the edge weights denote the distances. The problem is to find the shortest path from s to all nodes.



Problem 1: (small) Which algorithm would you use to solve the above problem? ■

Problem 2: (medium) Use the algorithm from previous question to solve the problem. Give sufficient details so that the solution method can be followed. ■

Critical path problem

A group of students following the Network Optimization course are solving the Project Assignment 2016. The assignment can be split into the following tasks:

Activity	Description	Predecessors	Duration
A	answer theoretical questions	E,B	5
B	buy color pencils		2
C	write theoretical model	E,B,G	7
D	write report	A,C	6
E	buy cake		4
F	install CPLEX	G	3
G	write up all paths in model		4
H	solve LP model	F	7

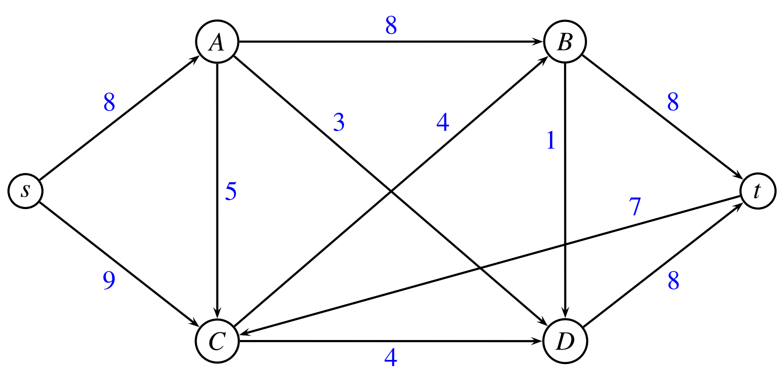
The duration of each task is given in the right-most column. A task cannot be started before the predecessor tasks have finished. Tasks can be carried out in parallel, by splitting the jobs between the group members.

Problem 3: (small) Order the tasks in topological order. ■

Problem 4: (medium) For each activity, find the earliest starting time and the latest starting time. Mark the critical path. ■

Maximum flow problem

Consider the following maximum flow problem where edge weights denote the capacity of an edge.

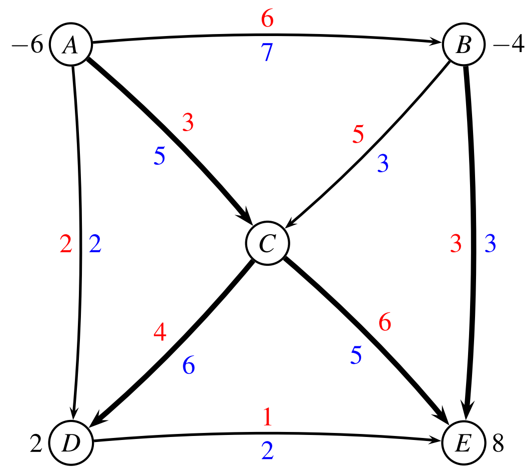


Problem 5: (medium) Solve the above maximum flow problem using an augmenting path algorithm. In each iteration use the augmenting path having *largest possible capacity*. Report which augmenting paths were used and their capacity. ■

Problem 6: (small) When the algorithm terminates, show the residual graph, indicate a minimum cut, and report the capacity of the cut. ■

Minimum cost flow

Consider the following network with edge costs c_{ij} (red) and edge capacities u_{ij} (blue). The demand b_i (black) in every node is positive for a sink, and negative for a source.



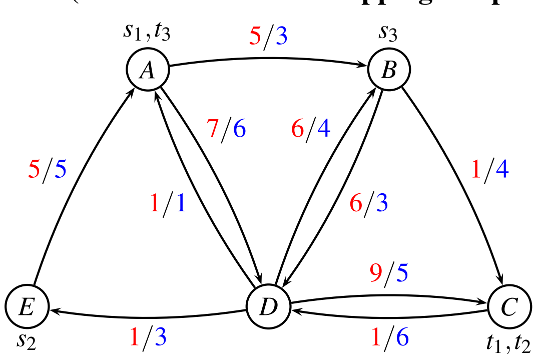
Having solved a maximum flow problem to find a feasible solution we are told that the tree solution T is defined by the bold edges in the above figure. Moreover, $L = \{(A,B)\}$ and $U = \{(A,D),(B,C),(D,E)\}$. The flow on edges L is at the lower bound, while the flow on edges in U is at the upper bound.

Problem 7: (small) Calculate the flow corresponding to the above feasible solution defined by T, L, U. ■

Problem 8: (large) Use the *network simplex* algorithm to solve the minimum cost flow problem to optimality. In each iteration report the tree solution, potentials on nodes, and reduced costs of edges. In the last iteration explain the stop criteria. ■

Column generation

MEDITLINE **(Route net of a liner shipping company)**



k	s_k	t_k	d_k
1	A	C	3
2	E	C	1
3	B	A	3

For edge (i,j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}. Cities are A = Alexandria, B = Barcelona, C = Casablanca, D = Dubrovnik, E = Ephesus.

The above instance of MEDITLINEis the same as in the Project Assignment 2016, except that the demand for commodity $k = 3$ has been changed to $d_k = 3$. Costs of flowing a container is indicated with red, and capacity of an edge is indicated with blue.

This means that all paths you found in the Project Assignment are the same:

number	commodity k	route r	d_k	$\sum_{(ij)\in r} c_{ij}$	$d_k \sum_{(ij)\in r} c_{ij}$
1	1	ABC	3	6	18
2	1	ABDC	3	20	60
3	1	ADBC	3	14	42
4	1	ACD	3	16	48
5	2	EABC	1	11	11
6	2	EABDC	1	25	25
7	2	EADBC	1	19	19
8	2	EADC	1	21	21
9	3	BCDEA	3	8	24
10	3	BDEA	3	12	36

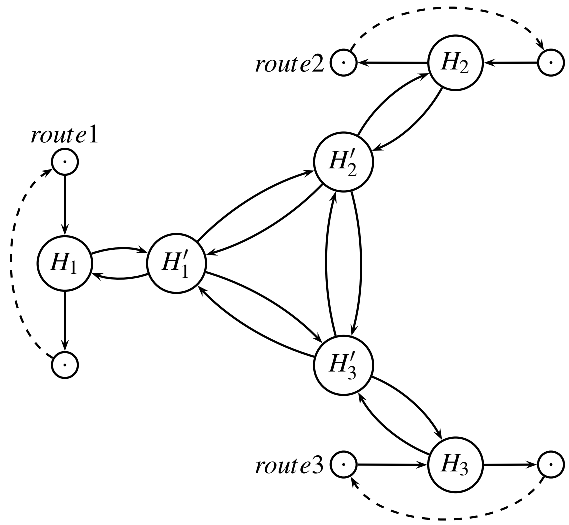
We solve the modified instance of MEDITLINE as an *unsplittable* multi-commodity flow problem through delayed column generation. After a number of iterations we have the following restricted master problem:

$$
\begin{array}{llllllll}
\min & 18x_1 & + 48x_4 & + 11x_5 & + 25x_6 & + 36x_{10} & & \\
\text{s.t.} & 3x_1 & & + 1x_5 & + 1x_6 & & \leq 3 & (y_{AB}) \\
 & & 3x_4 & & & & \leq 6 & (y_{AD}) \\
 & 3x_1 & & + 1x_5 & & & \leq 4 & (y_{BC}) \\
 & & & & + 1x_6 & + 3x_{10} & \leq 3 & (y_{BD}) \\
 & & & & & & \leq 6 & (y_{CD}) \\
 & & & & & & \leq 1 & (y_{DA}) \\
 & & & & & & \leq 4 & (y_{DB}) \\
 & & 3x_4 & & + 1x_6 & & \leq 5 & (y_{DC}) \\
 & & & & & 3x_{10} & \leq 3 & (y_{DE}) \\
 & & & 1x_5 & & + 3x_{10} & \leq 5 & (y_{EA}) \\
 & & & & & & & \\
 & x_1 + & x_4 & & & & \geq 1 & (\overline{y}_1) \\
 & & & x_5 + & x_6 & & \geq 1 & (\overline{y}_2) \\
 & & & & & x_{10} & \geq 1 & (\overline{y}_3) \\
 & & & & & & & \\
 & \multicolumn{7}{l}{x_1, x_4, x_5, x_6, x_{10} \geq 0}
\end{array}
\qquad (1)
$$

Solving the above LP-model we find the dual solution to (1) as $y_{AB} = -10$, while all other $y_{ij} = 0$. Moreover, we have $\overline{y}_1 = 48$, $\overline{y}_2 = 21$, $\overline{y}_3 = 36$.

Problem 9: (large) Calculate the reduced costs of all paths that are not used in model (1) and indicate which column should be added to the restricted master problem. ■

Modeling a hub



Consider Model 5 from the Project Assignment 2016. The graph is used to model a hub port, where three routes meet and transship containers between the vessels. We model the hub port H by using an extra node H_1, H_2 and H_3 for each route, and additional nodes H_1', H_2' and H_3'. Edges are added between all pairs of nodes H_1', H_2', and H_3'.

In this problem we are *only considering the crane capacities*. The crane capacity for loading/unloading is $u = 8$ containers for routes 1 and 2, while the capacity for loading/unloading is $u = 5$ containers for route 3 since it is operated by a smaller crane.

Problem 10: (medium) In the Project Assignment there were separate cranes for loading and unloading, each having capacity u. Assume now that loading and unloading is made by the *same* crane, having overall capacity u for both loading and unlaoding. How would you modify the graph to take this into account? The resulting graph should still be a multi-commodity flow problem. ■

Census rounding

You are consulting for an environmental statistics firm. They collect statistics and publish the collected data in a book. The statistics are about populations of different regions in the world and are recorded in multiples of one million. Examples of such statistics would look like the following table.

Table 1

Country	A	B	C	Total
grown-up men	11,6	9,4	3,0	24,0
grown-up women	12,7	10,5	3,8	27,0
children	1,7	2,1	0,2	4,0
total	26,0	22,0	7,0	55,0

We will assume here for simplicity that our data is such that all row and column sums are integers. The Census Rounding Problem is to round all data to integers without changing any row or column sum. Each fractional number can be rounded either up or down. For example, a good rounding for our table data would be as follows.

Table 2

Country	A	B	C	Total
grown-up men	11	10	3	24
grown-up women	13	10	4	27
children	2	2	0	4
total	26	22	7	55

Problem 11: (large) Consider first the special case when all data are between 0 and 1. So you have a matrix of fractional numbers between 0 and 1 as illustrated in the Table 3, and your problem is to round each fraction that is between 0 and 1 to either 0 or 1 without changing the row or column sums. Formulate the rounding problem in Table 3 as a network problem, and explain how a solution should be used to define a rounding of the matrix. Moreover, explain that the solution always will be integral. (You do not need to solve the specific problem). ■

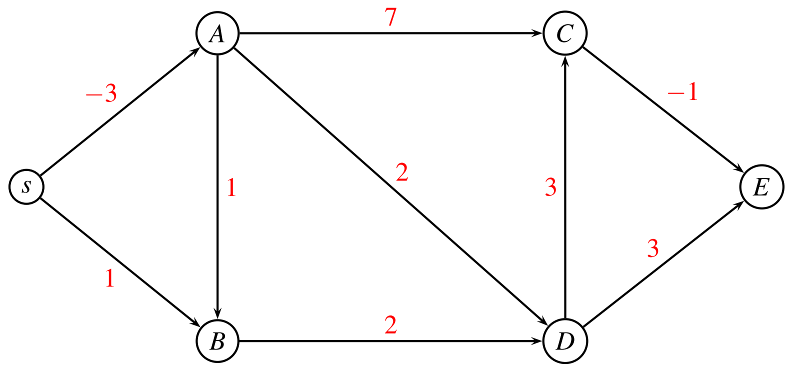
Table 3

Country	A	B	C	Total
grown-up men	0,1	0,7	0,2	1,0
grown-up women	0,4	0,6	0,0	1,0
children	0,5	0,7	0,8	2,0
total	1,0	2,0	1,0	4,0

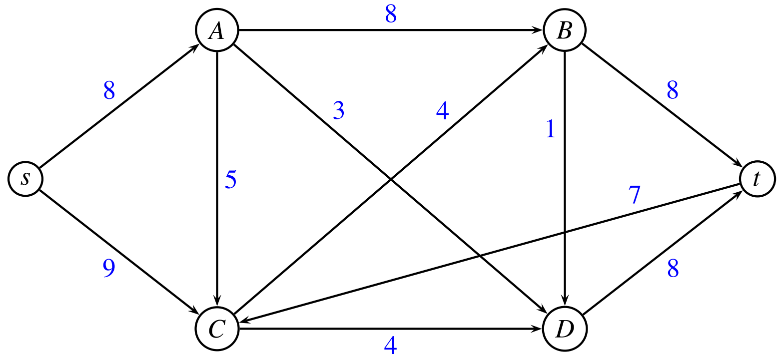
Problem 12: (medium) Consider now the general Census Rounding Problem as defined above, where row and column sums are integers, and you want to round each fractional number a to either $\lfloor a \rfloor$ or $\lceil a \rceil$. Formulate the rounding problem in Table 1 as a network problem, and explain how the solution should be used to define a rounding of the matrix. Moreover, explain that the solution always will be integral. (You do not need to solve the specific problem). ■

end of assignment

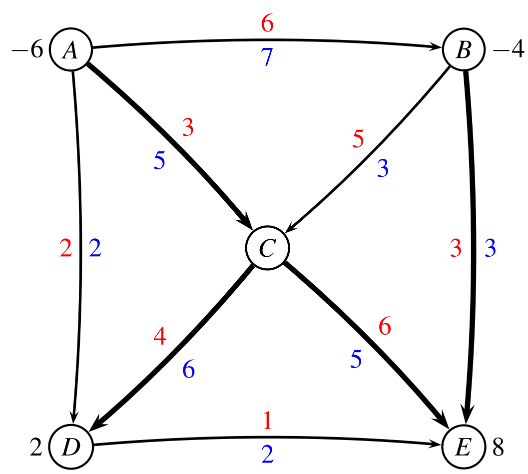
Shortest path



Maximum flow



Minimum cost flow



Modeling a hub